Weather Station

Student sheet

Name: _____ **Date:** _____

Safety: The box is not waterproof: never take it out in the rain; only an adult opens the windows.

1. What I see on the screen

Five cards: **Temperature**, **Humidity**, **Pressure**, **Altitude**, **Dew point**. You save with **Export data**, at the top right. The **Analysis** tab shows the pressure reduced to sea level and the **tendency over 3 hours**: steady, rising, falling, rapid. **The forecast is read from the trend, not from the value.** An ordinary tendency is only 0.1–0.3 hPa per hour. **Wait 10 min** after any move. Do not breathe into the box.

2. What I do on my own

Station: _____ **Box:** _____ **School's altitude:** _____ m

First I predict, before the first day. I do not change the prediction afterwards. 1. Before it rains, the pressure will ☐ rise ☐ fall ☐ stay the same, because _____

2. The humidity in the classroom will be ☐ bigger ☐ smaller than outdoors. **3.** I will get _____ forecasts out of 10 right.

One day's pressure chart (glued in or drawn here). Mark the steepest section. Pressure ☐ ↑ ☐ ↓ ☐ → *(rising, falling, steady)*

Predicts: _____

Daily recording sheet. One row per reading, at fixed times. **The room's state is written down with the number**, not later.

Date	**Time**	T **(°C)**	**RH (%)**	P **(hPa)**	T_d **(°C)**	**Room state (windows, heating, how many people)**

The tendency. Example: $1006.2 - 1008.4 = -2.2$ hPa in 3 hours, so $-2.2/3 = -0.73$ hPa/h — a clear fall, so cloud and rain.
Mine: _____ − _____ = _____ hPa in ____ hours, so _____ hPa/h.

Forecast log. The forecast is written *before* seeing the next day. The actual weather is recorded as it was, using the criterion fixed on day one.

Date	Tendency (hPa/h)	My forecast for tomorrow	Actual weather	Correct?

Correct: _____ out of _____ "Tomorrow like today": _____ out of _____ Did I beat it? ☐ YES ☐ NO

3. The conclusion I write

I predicted that, and the data show that _____

_____

The relative humidity was very different, but the dew point was not, because _____

_____

The 1013 hPa threshold does not work at our school because _____

_____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

_____